

LEBANESE UNIVERSITY

Faculty of Engineering III
Sensors & Instrumentation Project

Individual Contribution Report

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Project: Smart Building Security and Fire-Protection System

Contribution Summary

Throughout the development of the Smart Building prototype, I was actively involved in all group working sessions, collaborative planning, and execution phases alongside my teammates. My primary individual responsibility centered on the structural and physical realization of the system.

Specifically, I took the lead in designing and constructing the physical sample house that served as the operational framework for the entire project. Applying 3D mechanical design and drafting techniques, I planned and built a structure accurately dimensioned to accommodate the various hardware components and subsystems. This involved constructing a secure entry platform specifically designed to house the HX711 load cell and integrating the TCRT5000 infrared sensor for the dual-confirmation security gate. I also ensured the physical layout safely separated the hydraulic path—including the water container, mini pump, and YF-S201 flow sensor—from the low-voltage control electronics to mitigate any leak risks.

Beyond the physical modeling, I participated in the comprehensive hardware assembly and wiring sessions at the university. I collaborated with the team to mount the environmental actuators, such as the ventilation fan and the SG90 servo door mechanism, ensuring their mechanical movements and alignments operated smoothly within the sample house.

Furthermore, I was present for all major team meetings and testing activities. I actively assisted in running the integrated experimental campaign, observing the physical system responses—such as

the door actuation and fire-protection pumping sequences—to help the team collect the required multi-sensor data samples and validate the automatic responses.

Declaration

I certify that the activities described above accurately reflect my individual contribution to the Smart Building Security and Fire-Protection System project.

Signature: Nassim Amer